1.2 a)

Diameter (mm)	Area (cm^2)
25	4.91
50	19.64
75	44.18
100	78.54
125	122.72
150	176.71
200	314.16
300	706.86
450	1590.43

b) Dividing wafer area by die area, approximately 159,043 1mm $\times$ 1mm dice.

c) Dividing wafer area by die area,

$$\frac{159,043.13\text{mm}^2}{625\text{mm}^2} = 254.47,$$

so approximately 254 dice.

1.5 Substituting $Y = 2020$ into the given formula,

$$N(2020) = 1027 \times 10^{0.1505(2020-1970)} = 3.44 \times 10^{10},$$

around 34 billion transistors per microprocessor.

a) Using the given formula,

$$N(Y) = 2N(Y) \Rightarrow Y = \frac{\log_{10}(2)}{0.1505} = 2.002,$$

and thus the transistor count on a microprocessor would be expected to double about every two years.

b) Using the same formula,

$$Y = \frac{\log_{10}(10)}{0.1505} = 6.6445,$$

and thus the transistor count on a microprocessor would be expected to increase by a factor of 10 roughly every six and half years.

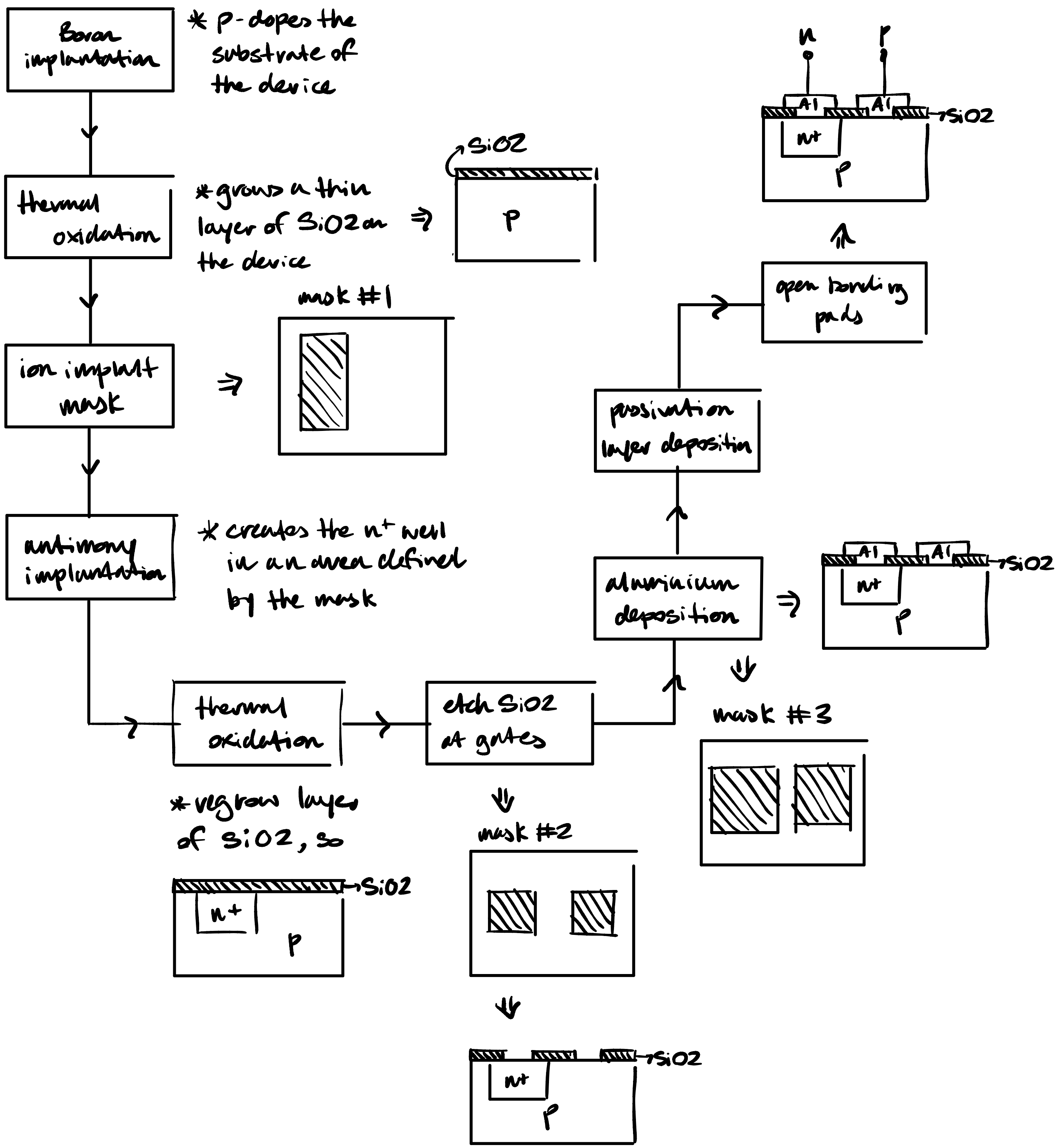
1.12 a) Using Fig 1.1(b), we may assume that approximately 175 10 $\times$ 10mm dice are on a 150mm wafer. Assuming 35% of the dice are good, this leaves us with $175 * 0.35 = 61.25$, so approximately 61 good dice. Assuming a cost per wafer of \$1,000, the cost per die is approximately

$$\frac{\$1000}{61} = \$16.39.$$

b) Again using Fig 1.1(b), we may assume that approximately 314 10×10mm dice are on a 200mm wafer. Using the same perecentage of good dice as above, this leaves $314 * 0.35 = 109.90$, or approximately 109 good dice. Then, approximate cost per wafer is

$$\frac{\$1000}{109} = \$9.17.$$

1.14 See below for included flowchart:



Boron implantation
* p-dopes the substrate of the device
thermal oxidation
* grows a thin layer of SiO2 on the device
SiO2
p
ion implant mask
mask #1
antimony implantation
* creates the n+ well in an area defined by the mask
thermal oxidation
* regrow layer of SiO2, so
SiO2
n+
p
etch SiO2 at gates
mask #2
SiO2
n+
p
aluminium deposition
mask #3
Al
Al
SiO2
n+
p
passivation layer deposition
open bonding pads
n
p
Al
Al
SiO2
n+
p